Thermal conductivity by Lee's methord

Aim

To measure the thermal conductivity of bad conductors of heat.

Analysis

Newton's law of cooling is given by

$$\frac{dT}{dt} = -b(T - T_a)$$

For solving this differential equation we shall put $x = T - T_a$

$$\begin{aligned}
\frac{dx}{dt} &= -bx \\
\int \frac{dx}{x} &= \int -bdt \\
\ln|x| &= -bt + c \\
x &= e^c e^{-bt} \\
T &= ae^{-bt} + T_a
\end{aligned}$$

We see that the constant a is independent from everything else as it is derived from the constant of integration, which does not depend on the data. Performing fitting for the model

$$y = ae^{-bt} + c$$

We obtain the coefficients as listed in the table below.

Material	a	b	c	R^2
Glass	21.42	0.637	67.93	0.9966
Ebonite	23.48	0.257	62.40	0.9962
Wood	24.48	0.115	54.13	0.9971

Figure 1: Glass

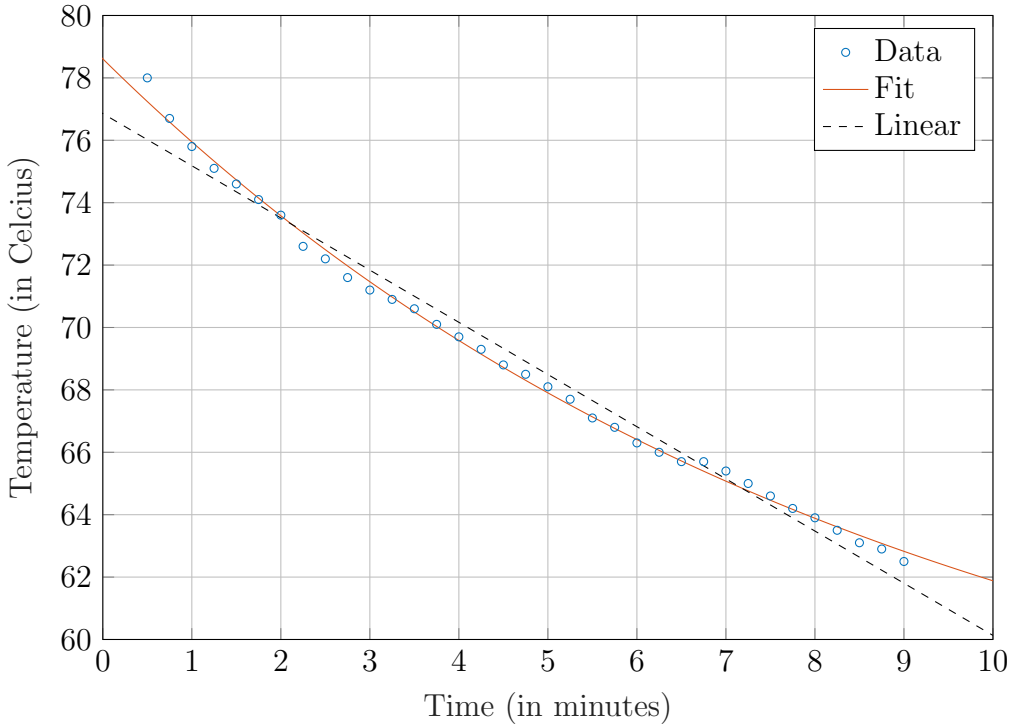


Figure 2: Ebonite

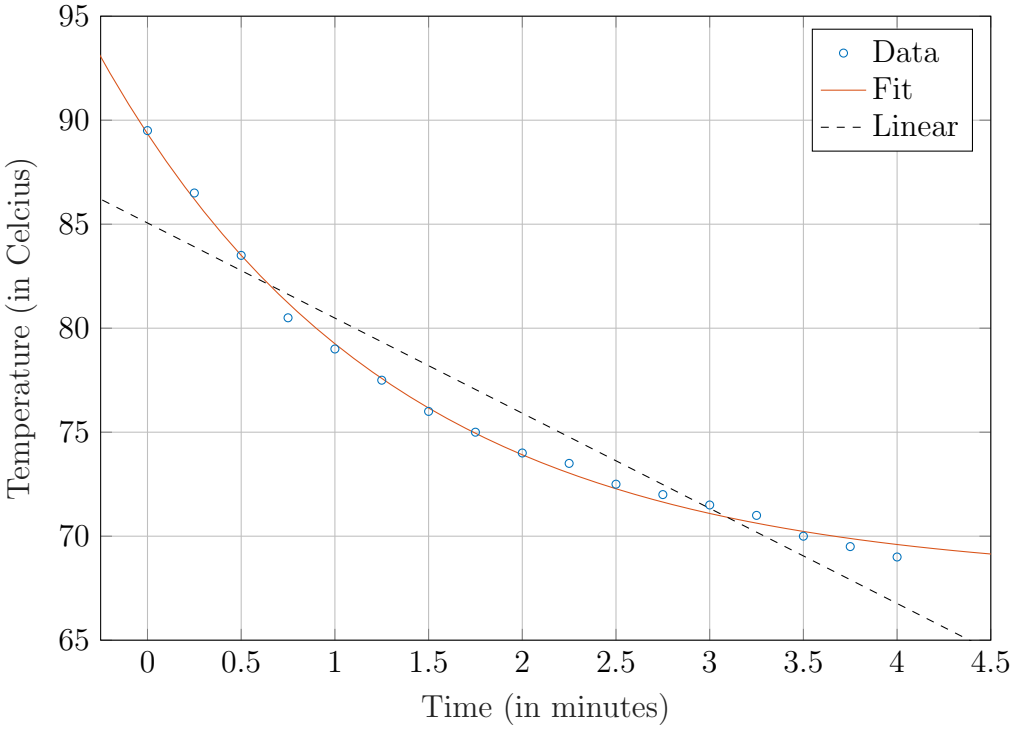
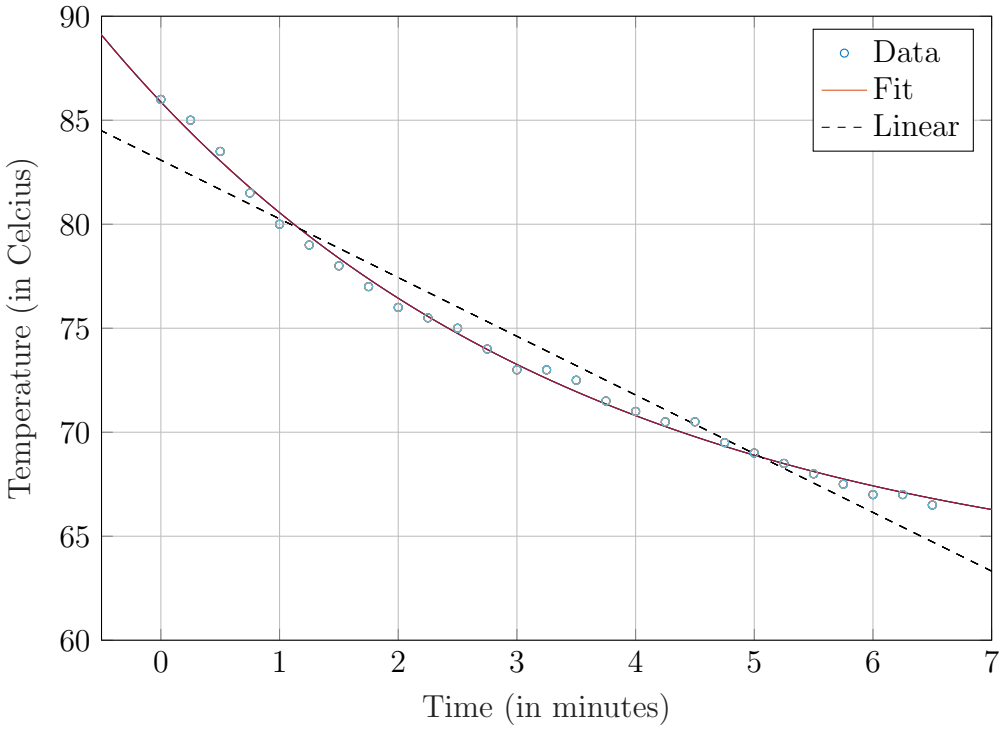
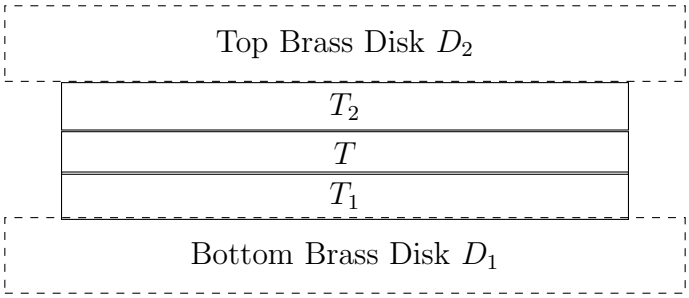


Figure 3: Wood

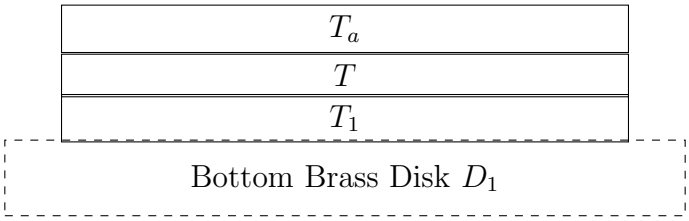


We see that the values for a is consistent. However for the values of c we are getting it around $60°C$. This is not the ambient temprature. The following section explains the values observed.

For the first part of the experiment, when the sample disk is at equilibrium with the two brass disks, its temprature profile is a gradient like the diagram shown below(not to scale).

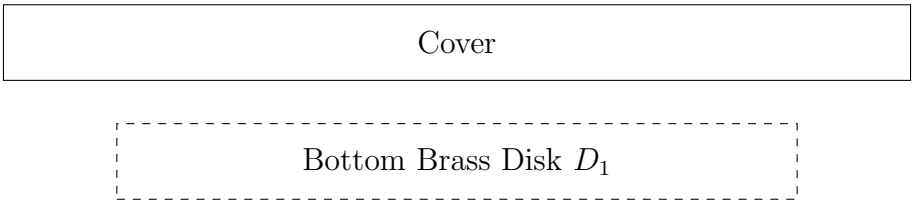


The surface is cooled to room temprature but the insides are still hot because it is a bad conductor. Thus, during the second part of the experiment the temprature profile looks like the diagram shown below.

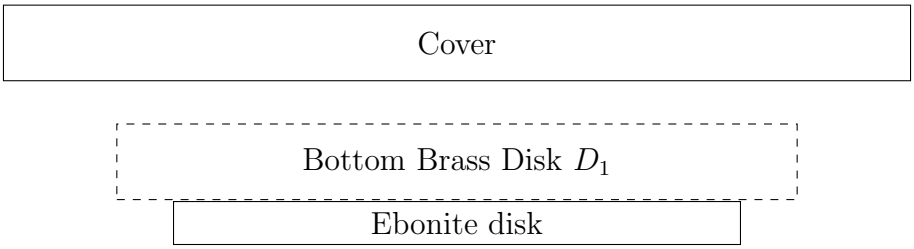


Having a thin disk helps us to approximate the temprature profile to a simpler model in addition to not waiting long for the system to achieve equilibrium.

Thus, during the second part of the experiment, the disk has cooled a bit but not as much to reach room temprature. We can conclude that the value of c is the averaged value of the temprature of sample disk and the air below D_1 across the second experiment. However, for smaller intervals, we can approximate the curves to a straight lines at the tempratures $T_1 \pm 7°C$ as seen in the plots. In order to reduce the amount of error due to losses in heat due to convection, I repeated the experiment with the following changes. First, I put a cover over the the cooling disk a few centimeters above it and then measured the cooling rate as shown in the diagram below.



Measuring the cooling rate gave the value of $2°C$ per minute at around $75°C$. Then, I put the ebonite disk below the brass disk



This would give the value of 1.992 around the same temprature range. This shows that the effect of the disk is negligible due to the sample disks being bad conductors and the value of k would come out

Using these values we can proceed with the calculations.

$$ms\frac{dT}{dt} = k\frac{A}{l}(T_2 - T_1)$$

rearanging terms we get

$$k = \frac{msl}{\pi r^2(T_2 - T_1)}\frac{dT}{dt}$$

Calculating for the different materials, we get the values

$$\begin{aligned} \text{Glass} &: 0.336 \\ \text{Ebonite} &: 0.275 \\ \text{Wood} &: 0.160 \end{aligned}$$

We ge the following values for error

$$\begin{aligned}\sigma_{\Delta T} &= 1^{\circ}C \\ \sigma_l &= 0.1cm \\ \sigma_r &= 0.02mm\end{aligned}$$

There is another error term that is introduced in the slope by the fitting by checking how much the value changes if we vary the temprature range over which k is measured(due to different materials). This gives the following error in slope

$$\sigma_{slope} = 0.2$$

If f is a function of x_is then the uncertainty in f is given by

$$\sigma_f^2 = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \sigma_{x_i} \right)^2$$

k is a function of $\sigma_{\Delta T}$, σ_l, σ_r and σ_{slope}

$$\left(\frac{\partial f}{\partial l} \sigma_l \right)^2 + \left(\frac{\partial f}{\partial r} \sigma_r \right)^2 \approx 0$$

$$\sigma_k^2 = \left(\frac{msl}{\pi r^2 (T_2 - T_1)^2} \frac{dT}{dt} \sigma_{\Delta T} \right)^2 + \left(\frac{msl}{\pi r^2 (T_2 - T_1)} \sigma_{slope} \right)^2$$

which gives the error in k

$$\begin{aligned}\text{Glass} &: \sigma_k = 0.040 \\ \text{Ebonite} &: \sigma_k = 0.039 \\ \text{Wood} &: \sigma_k = 0.018\end{aligned}$$

Result

We have found the conductivity of the materials and thier values are listed below.

$$\begin{aligned}\text{Glass} &= 0.336 \pm 40 \times 10^{-3} W/mK \\ \text{Ebonite} &= 275 \pm 39 \times 10^{-3} W/mK \\ \text{Wood} &= 160 \pm 18 \times 10^{-3} W/mK\end{aligned}$$